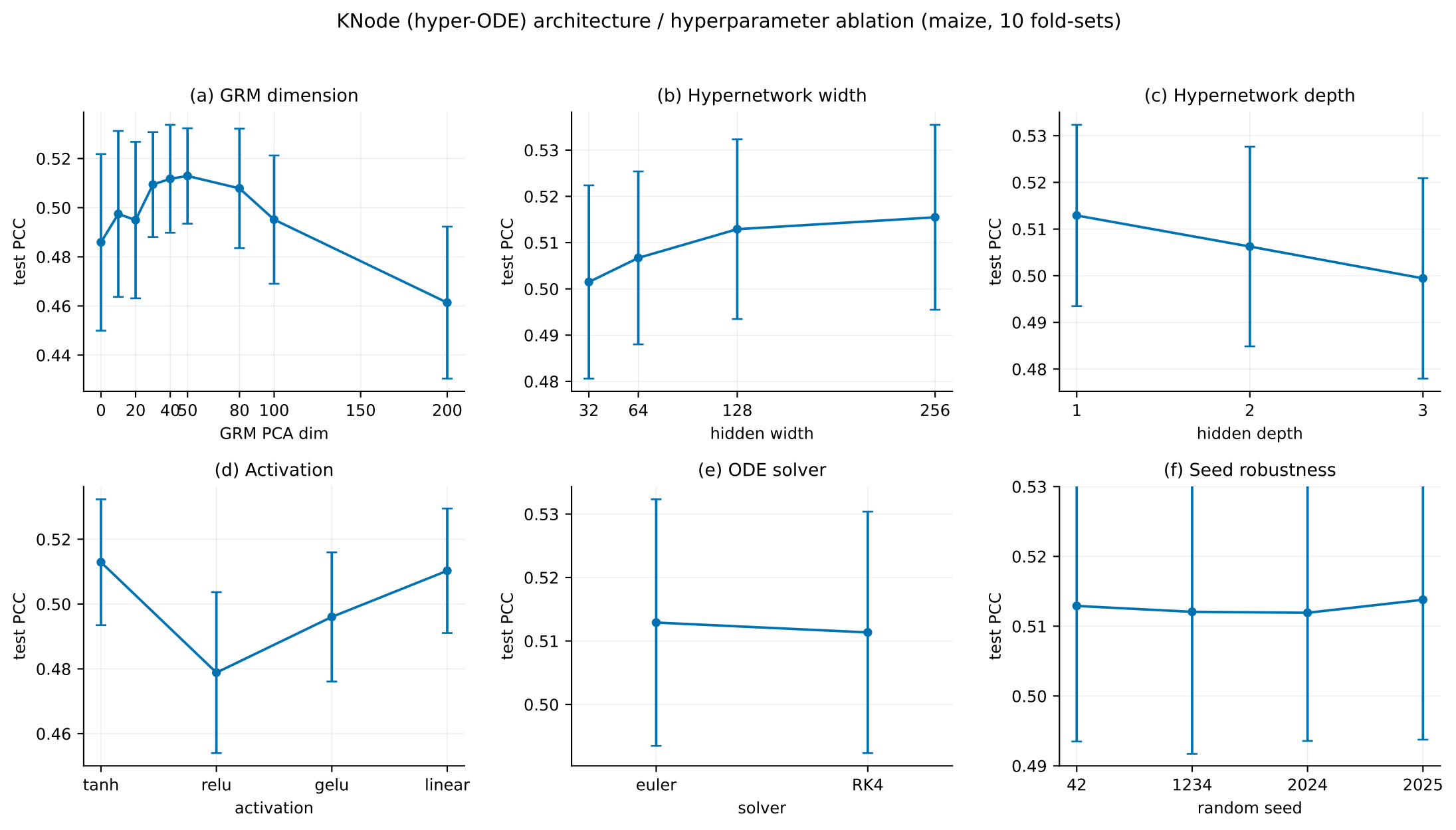
KNode (hyper-ODE) architecture / hyperparameter ablation (maize, 10 fold-sets)
(a) GRM dimension
test PCC
0.52
0.50
0.48
0.46
0.44
0
20
40
50
80
100
150
200
GRM PCA dim
(b) Hypernetwork width
test PCC
0.53
0.52
0.51
0.50
0.49
0.48
32
64
128
256
hidden width
(c) Hypernetwork depth
test PCC
0.53
0.52
0.51
0.50
0.49
0.48
1
2
3
hidden depth
(d) Activation
test PCC
0.52
0.50
0.48
0.46
tanh
relu
gelu
linear
activation
(e) ODE solver
test PCC
0.53
0.52
0.51
0.50
euler
RK4
solver
(f) Seed robustness
test PCC
0.53
0.52
0.51
0.50
0.49
42
1234
2024
2025
random seed